STAGE	AS-IS · current investigation	TO-BE · assisted investigation	BENEFITS · simulated
DETECT **0.5 day as-is** 0.1 day to-be **simulated**	MANUAL **Alarm noticed on the MX-400 etch.** The operator tells a technician in person. 1 manual hand-off, simulated.	**Statistical process control opens the ticket.** The console shows the alert. Nobody retypes it. 0 manual hand-offs, simulated.	**Time to root cause** 4.0 days down to 1.2 days, simulated. Detect through Diagnose only. Approval days are not reduced.
TRIAGE **0.5 day as-is** 0.2 day to-be **simulated**	MANUAL **The technician types the ticket into email.** The quality engineer assigns an owner by reply. 2 manual hand-offs, simulated.	HUMAN AGENT **The engineer opens the ticket.** The agent plans the pack and does not diagnose. 1 human hand-off, simulated.	**Fewer hand-offs** 13 manual hand-offs down to 2, simulated. The engineer opens the ticket. The engineer approves.
GATHER EVIDENCE **2.0 days as-is** 0.4 day to-be **simulated**	MANUAL **Run sheet, historian, control chart, change log.** Prior cases and supplier lots. A meeting compares them. 5 manual hand-offs, simulated.	AGENT **Read-only calls. One evidence pack.** One plain sentence on each evidence item. 0 manual hand-offs, simulated.	**Traceable evidence trail** One pack, with a citation on each claim. The mail thread is not the record. Simulated comparison.
DIAGNOSE **1.0 day as-is** 0.5 day to-be **simulated**	MANUAL **The process engineer states the last known cause.** The four cause classes are not checked as a set. 1 manual hand-off, simulated.	AGENT **Four causes are checked on every incident.** The agent ranks one and names what rules it out. 0 manual hand-offs, simulated.	**Consistent four-cause check** Engineering change, out of spec, supply-chain material, and process drift. The same four, every time.
DECIDE / APPROVE **2.0 days as-is** 2.0 days to-be **simulated**	MANUAL **Approval is a meeting or a mail reply.** No evidence pack is attached to the ask. 2 manual hand-offs, simulated.	HUMAN **Grounding rejects a claim with no citation.** The engineer approves with a signed token. 1 human hand-off, simulated.	**The human stays the approver** A signed token is required. Agents do not write the record. The person decides.
CAPA **62 days as-is** 40.3 days to-be **simulated**	MANUAL **The record is re-keyed into the quality system.** Management signs it and does not see the files. 2 manual hand-offs, simulated.	**The action service writes the record.** The agent draft is not a write. 0 manual hand-offs, simulated.	

Glossary

Short forms on this sheet. All figures are simulated.

CAPA corrective and preventive action

Detect through Diagnose is the time to a hypothesis: 4.0 days down to 1.2 days, simulated.

The CAPA row is the record cycle, simulated. Decide / approve stays at 2.0 days, simulated.

LEGEND

- As-is step. People, email, and files.
- Red MANUAL badge. A hand-off.
- To-be rules. Code opens the ticket and writes.
- To-be agent. Plans, gathers, and ranks.
- To-be human. Opens the ticket and approves.
- Benefit. Every figure is simulated.

MERIDIAN PRECISION SYSTEMS
Quality and Engineering Issue Resolution

Current and future state

G5 · Issue #74 Version v1.0 Date 2026-10-14

Prepared by StatusNeo

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